Curriculum Vitae of

Md Mutasim Billah, Ph.D.

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Academic Appointment

- **Visiting Assistant Professor** Fall 2025–Present
  - Department of Mathematics, Statistics, and Computer Science (MSCS)
  - **Macalester College**, St. Paul, Minnesota

Education

- **Ph.D.** in Statistics Fall 2023–Summer 2025
  - **Michigan Technological University**, Houghton, Michigan
  - **Dissertation**: "Methods in Statistics, Machine Learning, and Deep Learning for Combining Multi-Omics Datasets"
  - **Advisor**: Dr. Kui Zhang, Professor, Dept. of Mathematical Sciences
- **M.Sc.** in Statistics Fall 2019–Fall 2023
  - **Michigan Technological University**, Houghton, Michigan
- **Graduate Studies** in Applied Statistics Fall 2018–Summer 2019
  - **University of Northern Colorado**, Greeley, Colorado
- **M.Sc.** in Statistics, Biostatistics & Informatics Fall 2012–Summer 2013
  - **University of Dhaka**, Dhaka, Bangladesh
- **B.Sc.** in Statistics, Biostatistics & Informatics Fall 2007–Summer 2012
  - **University of Dhaka**, Dhaka, Bangladesh

Honors and Awards

- **Finishing Fellowship** from Michigan Technological University in Summer 2025
- **Recognition Award** for exceptional average of 7 dimensions in teaching in Fall 2020
  - Provost's office for academic affairs, Michigan Technological University
  - Only 73 instructors (86 sections out of more than 1,000 evaluated) university-wide were rated this highly by students
- **Recognition Award** for outstanding teaching in Spring 2020
  - Provost's Office for Academic Affairs, Michigan Technological University
- **Graduate Course Excellence Awards**, Mathematical Sciences, Michigan Technological University
  - Awarded for top performance in Ph.D.-level coursework, including Advanced Topics in Statistics (Probability I (Fall 21), Probability II (Spring 22), Mathematical Statistics I (Fall 19, Fall 20), Mathematical Statistics II (Spring 20, Spring 21), Linear Algebra (Fall 19), and Categorical Data Analysis (Spring 20).

Teaching Experience

- **Instructor of Record**

  Visiting Assistant Professor Fall 2025–Present
  Department of MSCS, Macalester College
  - Instructing courses at both the foundational and advanced levels of the undergraduate statistics and data science curriculum

  Graduate Teaching Instructor Spring 2020–Spring 2025
  Department of Mathematical Sciences, Michigan Technological University
  - Instructed MA 3710: Engineering Statistics across multiple modalities: in-person, hybrid, synchronous online, and asynchronous online

- Maintained average student evaluation scores of **4.21+/5.00** in **14 of 15 semesters** (~**93%** of courses). Notably, **4.21** is the overall average student evaluation score of MTU faculty over the past seven years

Graduate Teaching Instructor Fall 2018–Summer 2019
Department of Applied Statistics and Research Methods, University of Northern Colorado
- Instructed Introduction to statistics, a required general-education course

- **Teaching and Learning Support**

Graduate Teaching Assistant
Dr. Kui Zhang, Mathematical Sciences, Michigan Technological University Fall 2022–Spring 2023
Beth Reed, Mathematical Sciences, Michigan Technological University Fall 2019

- **Instructor of records/Co-instruction/Teaching Assistant course experience**

Course	**Institution**	**Semester**
Introduction to Statistical Modeling	Macalester College	Fall 2025• (2 sections) Spring 2026•• (2 sections)
Statistical Machine Learning	Macalester College	Spring 2026••
Engineering Statistics	Michigan Technological University (Evaluations are available upon request)	Spring 2025• Summer 2024† Spring 2024† Fall 2023† Summer 2023† Spring 2023† Summer 2022† Spring 2022† Fall 2021† Summer 2021† Spring 2021† Fall 2020†† Summer 2020†† Spring 2020• Fall 2019**
Applied Generalized Linear Models	Michigan Technological University	Spring 2023*
Design & Analysis of Experiments	Michigan Technological University	Fall 2022*
Statistical Methods	Michigan Technological University	Fall 2019**
Introduction to Statistical Analysis	University of Northern Colorado	Fall 2018• Spring 2019• Summer 2019•

• Instructor of record (in person)
•• Instructor of record (in person; scheduled)
† Instructor of record (asynchronous online)
†† Instructor of record (synchronous online)
* Teaching assistant (asynchronous online)
** Teaching assistant (in person)

Research Agenda

- **Research Interests**
  - **Core Methodology:** Developing methods in statistics, machine learning (ML), and deep learning (DL) tailored to genomic data; Statistical frameworks for multi-omics integration; Methodological advances in missing data imputation and uncertainty modeling for high-dimensional, incomplete biological data; Causal inference and modeling of biological heterogeneity

  - **Application Domain:** Dissection of gene regulatory architecture across diverse tissues (pleiotropy, cis/trans-effects); High-resolution fine-mapping of causal variants to elucidate disease etiology; Translation of multi-omics discoveries into predictive biomarkers and public-health interventions; Genomic medicine and health-equity research in ancestrally diverse and historically underserved populations
  - **Computational Science:** Scalable and reproducible computational genomics for cohort-scale studies (GWAS, GTEx, dbGaP); Development of end-to-end analysis pipelines for secure, controlled-access omics data; High-performance computing implementations that couple DL with statistical genomics for large-scale discovery

- **Active and Future Research Plan**
  - Extending previously developed multi-tissue TWAS frameworks (e.g., TWAS-CTL, G-Boost-CTL) to leverage GWAS summary-level data
  - Benchmarking advanced ML and DL Models for cross-tissue gene expression imputation
  - Investigating the distinct roles and interactions of local and distant genetic variants in controlling gene expression across different tissues within the TWAS framework
  - Developing advanced ML and DL pipelines to impute missing values in multi-tissue gene expression datasets (e.g., GTEx reference panel)
  - Developing sparsity-inducing, power-adaptive procedures to identify and remove non-informative tissues in multi-tissue TWAS
  - Developing unsupervised clustering to characterize pleiotropic effects in GWAS

Research Experience

- **Doctoral Researcher** Fall 2022–Summer 2025
  Department of Mathematical Sciences, Michigan Technological University
  - Developed TWAS-CTL, a novel and computationally efficient multi-tissue learning method that enhances statistical power by strategically weighting and integrating gene expression data from multiple learners, outperforming existing benchmarks
  - Introduced G-Boost-CTL, an advanced framework that further improves TWAS-CTL by incorporating a dual-weighting scheme that uses information from the GWAS cohort itself to automatically emphasize tissues carrying the relevant genetic signals, delivering significant gains in statistical power and discovering more biologically plausible disease loci while preserving error control compared to other established methods in multi-tissue TWAS
  - Demonstrated significant gains in discovery power and identifying more causal genes by replacing traditional linear models in the multi-tissue TWAS pipeline with non-linear machine learning and deep learning engines (gradient-boosted trees and deep neural networks) to capture complex, heterogeneous gene regulatory patterns, outshining conventional benchmarks
- **Graduate Research Assistant** Summers 2023, 2024
  Dr. Kui Zhang, Professor, Department of Mathematical Sciences, Michigan Technological University
  - Secured NIH dbGaP approval and coordinated the download of > 2 TB of protected GTEx genotype and RNA-seq data; built automated Bash/R pipelines for de-identification, quality control, and harmonization across 49 tissues
  - Designed and coded scalable R-/Python-based workflows for both established (PrediXcan, UTMOST) and new cross-tissue TWAS methods (TWAS-CTL, G-Boost-CTL)
  - Conducted large-scale power simulations, statistical validation, and data-visualization that underpin three forthcoming manuscripts
- **Statistical Consultant, Research Consulting Lab** Fall 2018–Summer 2019
  Department of Applied Statistics and Research Methods, University of Northern Colorado
  - Acted as a lead statistical consultant, overseeing client intake, aligning project needs with consultant expertise, and mentoring new team members while coordinating statistical training workshops
  - Delivered full-spectrum consulting support for qualitative, quantitative, mixed-methods, and evaluation studies – from research design and methodology development to final data analysis and reporting
  - Designed and executed rigorous statistical analysis plans tailored to diverse academic disciplines
  - Collaborated with graduate students, faculty, and staff on data analysis, research presentations, and thesis projects using R, SAS, and SPSS

Publications

- **Manuscripts Under Review**
  - **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[†] (2025). TWAS-CTL: A robust and efficient framework in multi-tissue transcriptome-wide association studies using cross-tissue learner.
    - Submitted to PLOS Genetics; Preprint available on bioRxiv: https://doi.org/10.1101/2025.10.25.684514
- **Manuscripts in Preparation**
  - GBoost-CTL: A novel method in multi-tissue transcriptome-wide association studies in cross-tissue learner incorporating GWAS information; **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[•]
    - Target journal: PLOS Genetics; Intended submission timeline: December 2025
  - Application of Deep Learning and Boosted Tree Methods in multi-tissue TWAS framework; **Billah, M.**, Acheampong, S., Kareem, K., Subedi, M., Silwal, S., Zhang, K.[†]
    - Target journal: PLOS ONE; Intended submission timeline: February 2026

[†] Professor, Michigan Technological University [•] Professor, University of Southern California

Presentations

- **Juried Poster Presentations**
  - **Billah, M.**, Wei, H., Sun, F., Zhang, K.; "GBoost-CTL: A novel method in multi-tissue transcriptome-wide association studies in cross-tissue learner incorporating GWAS information"–American Society of Human Genetics Annual Meeting 2025
  - **Billah, M.**, Wei, H., Sun, F., Zhang, K.; "TWAS-CTL: A robust and efficient framework in transcriptome-wide association studies using cross-tissue learner"–American Society of Human Genetics Annual Meeting 2024
- **Non-juried Poster Presentations**
  - **Billah, M. (Presenter)**; "A simple Analogy for Research Plan in Grant Application"–Prototype Poster Session, Associated Colleges of the Midwest (ACM) Grant Workshop 2025
  - **Billah, M. (Presenter)**, Acheampong, S., Kareem, K., Subedi, M., Silwal, S., Zhang, K.; "Application of Deep Learning and Boosted Tree Methods in multi-tissue TWAS Framework"–Graduate Research Poster Session, Michigan Technological University, Summer 2025
  - **Billah, M. (Presenter)**; "Analyzing dementia using Open Access Series of Imaging Studies data: a comparison between multiple linear regression and beta regression analysis approach"–Research Evening, University of Northern Colorado, Spring 2019
- **Non-Juried Podium Presentations**
  - **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[†]; "GBoost-CTL: A novel method in multi-tissue transcriptome-wide association studies in cross-tissue learner incorporating GWAS information"–Research Seminar, Mathematical Sciences, Spring 2025
  - **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[†]; "TWAS-CTL: A robust and efficient framework in transcriptome-wide association studies using cross-tissue learner"–Research Seminar, Mathematical Sciences, Spring 2024

Grants

- **Grants in Preparation**
  - "Application of Deep Learning methods and its improved accuracy in predicting multi-tissue gene expression from genotypes"–R03 NIH Grant
- **Travel Grants**
  - Faculty Travel and Research Grant, Macalester College; Amount: $1250 — Fall 2025–Summer 2026
  - Department of Mathematical Sciences Travel Grants, MTU; Amount: $1000 — Fall 2024
  - Graduate Student Government (GSG) Travel Grants, MTU; Amount: $500 — Fall 2024

Service and Leadership

- **Member, Transformation Committee, MSCS, Macalester College** — 2025–2026
  - Preceptor (GTA) Program Development & Training

  - Faculty Support & Mentoring
  - DEI Initiatives
- **Treasurer and Vice President (Elected), Muslim Student Association** Spring 2021–Spring 2023
  - Graduate Student Government (GSG), Michigan Technological University
- **President (Elected), Bangladesh Student Association** Spring 2020–Spring 2021
  - GSG, Michigan Technological University

Professional Development

- **Workshops**
  - Associated Colleges of the Midwest Grant Workshop, Macalester College August 12-15, 2025
- **Conferences**
  - Upper Peninsula's Teaching and Learning Conference, Bay College in Escanaba May 13-14, 2024
  - UP's Teaching and Learning Conference, Lake Superior State University, May 15-16, 2023
  - UP's Teaching and Learning Conference, Northern Michigan University May 24-25, 2022

Computational, Instructional, and Analytical Proficiency

- **Instructional Design and Teaching Tools**
  - **Pedagogical Frameworks:**
    - Adept in inclusive frameworks including **Universal Design for Learning (UDL)** and **Equitable Assessment** (e.g., Alternative Grading, Specifications Grading).
    - Skilled in student-centered active learning methodologies such as **Project-Based Learning**, **Inquiry-Based Learning**, and **Flipped Classroom** design.
  - **Computational Pedagogy:**
    - Specialized in teaching reproducible workflows by designing interactive course websites using **Quarto** and **R**, with version control managed via **GitHub**.
  - **Learning Technologies & Platforms:**
    - Proficient with Learning Management Systems (**Canvas**, **Moodle**) and a suite of instructional tools including **Panopto**, **iClicker**, **SpeedGrader**, and **WebAssign**.
- **Statistical Programming and Data Analysis**
  - Proficient in **R**, **Python**, **SAS**, **PLINK**, **GCTA**, and **SPSS**

References

- **Dr. Kui Zhang**
  - Professor, Mathematical Sciences, MTU
  - Email: kuiz@mtu.edu; Phone: 906-487-2918
- **Dr. Qiuying Sha**
  - Distinguished Professor, Mathematical Sciences, MTU
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